Meta-TTL: Meta-Learning Self-Improvement Policies for Language Agents

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HarnessRSI 69 ★★

evidence: improver

Paper

Code

1 What changes, what is kept

Inner loop: one TTL session on one task

6 episodes (Jericho) or 3 (WebArena-Lite, τ²-bench)

Actor (frozen)
Gemini-3-Flash plays episode k with prompt ρk

history →

← prompt

Meta-agent
GPT-5, steered by the **meta-prompt ϕ (learned)**

session score = W-AUC (later episodes weigh more)

environment resets each episode; only the prompt carries over

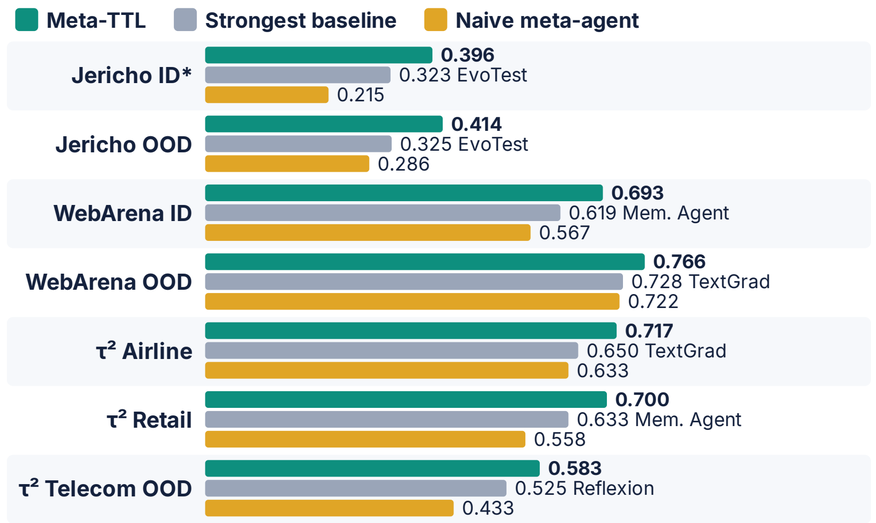
Outer loop: meta-training of ϕ (12–15 iterations)

1. **Run the expert ϕ⋆**
   sample a training task, run a full TTL session
2. **Two-stage reflection**
   extract task facts, then write candidate ϕ˜
3. **Validate both**
   expert and candidate on validation tasks; W-AUC → per-task z-score
4. **Keep rule**
   BIS = mean(z̃ − z⋆) over tasks; replace ϕ⋆ only if BIS > 0

205–350 sessions, about $30–70 per benchmark

then freeze ϕ⋆ and apply it to unseen tasks
weights never change; the kept artifact is text

2 Headline: best average W-AUC in 7 of 7 settings



Tables 1–3, W-AUC, Gemini-3-Flash actor. Runs: Jericho 5, WebArena 3, τ² not stated; no SDs. *Jericho ID = the meta-training games.

3 Controls that decide if the gain is real

Naive meta-agent (gold bars): a baseline with the same architecture and models but an unoptimised policy; below Meta-TTL in all 7 settings.

Table 5: Jericho W-AUC, Qwen3-4B meta-agent; ID = training games

Meta-agent training	ID avg	OOD avg	Temple
Untrained	0.146	0.168	0.129
GRPO (1,920 rollouts)	0.150	0.174	0.114
GEPA outer loop	0.152	0.198	**0.154**
Meta-TTL (350 rollouts)	**0.164**	**0.232**	0.129

Table 4: Qwen3-8B, K = 3; Meta-TTL reuses the main-run meta-prompt; ID = training games; not on the 0.396 scale

Static / SFT / GRPO	0.084 / 0.116 / 0.118	0.094 / 0.092 / 0.114
SFT+MAML	0.122	0.157
Meta-TTL	**0.182**	**0.201**

App. D (2nd backbone): Jericho and WebArena-Lite only, 3 baselines; Jericho OOD avg score 0.343 vs 0.334.

4 Where the gain is absent or thin

- **Jericho ID = the training games**; gap to EvoTest 0.162‡ Detective, 0.011‡ Zork 1, 0.046‡ Temple
- Ties/near-ties, no CIs: Zork 3 OOD 0.427 vs 0.423 (EvoTest); WebArena GitLab 0.883 = AWM; Shopping Admin 0.760 vs 0.749; τ² Retail avg score 0.633 = Reflexion
- Temple (ID, Table 5): 0.129 vs GEPA 0.154. OOD games, Table 4: Zork 3 SFT+MAML 0.405 vs 0.238; Balances GRPO 0.147 vs 0.131

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Reports: unseen-domain evaluation (Jericho ID = training games), Naive-meta-agent baseline, RL and GEPA comparison, cost, second backbone.

Does not report: per-cell SDs, matched-compute baselines, repeated meta-training, test-set sizes, leakage check.

Takeaway

A validation-gated, learned meta-prompt beats hand-designed TTL baselines and an un-optimised meta-agent in every reported average. The direction is consistent; the size is untested: no SDs, the Jericho ID rows are training games, several cells tie, and baselines get no matched meta-training budget.